

Changelog for GPW WATS 3.04 Rest API Messages for IDDS.

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2. clarification of the documentation content or removing ambiguity;
3. implementation of approved change requests or;
4. regulatory changes.

2. PREFACE

This section describes the basic information about GPW WATS Rest API Messages for IDDS. You can learn about the target audience, the document purpose and all the necessary documents you should read in relation to this Specification.

2.1. TARGET AUDIENCE

This document presents all adjustments that has been made in the newest release. The document is always compared with its previous version. Comparison of format changes is not included for readability purposes.

2.2. DOCUMENT'S PURPOSE

This document presents all adjustments that has been made in the newest release. The document is always compared with its previous version. Comparison of format changes is not included for readability purposes.

2.3. ASSOCIATED DOCUMENTS

Please check the following documents to learn more about IDDS:

- GPW WATS 3.02 Internet Data Distribution System
- GPW WATS 3.03 Streaming Messages for IDDS.

3. DOCUMENT HISTORY

Version	Date	Description
0.61	29.02.2024	Initial publication of the document.
0.62	25.03.2024	Endpoint structures have been updated accordingly to v0.62.

4. REFERENCE DATA ENDPOINTS

For each endpoint you need to attach Authorization token. You can find description how to receive and use the token in *GPW WATS 3.02 Internet Data Distribution System*.

The following **responses** apply for each endpoint:

All API calls produces the following media types according to the Accept request header; the media type is conveyed by the Content-Type response header.

Currently **accepted media types**:

- application/json

Expected **response codes**:

- 200 OK - returns data with specified type described for each endpoint in following chapters,
- 400 BadRequest - you send incorrect request (e.g., malformed request syntax, invalid request message framing),
- 401 Unauthorized - you must add valid authorization token to the header "Authorization: Bearer [token]",
- 403 Forbidden - you do not have an access rights to requested resource,
- 404 NotFound - data not found (check request URL).

4.1. GET MARKET STRUCTURE INFORMATION

```
get /api/v1/reference/{date}/market-structures
```

This endpoint retrieves market structure.

Path parameters

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date.

Return type

array[MarketStructureView]

Example data

Content-Type: application/json

```
[ {  
  "children" : [ null, null ],  
  "mic" : "mic",  
  "name" : "name",  
  "id" : 0,  
  "mic": "string",  
  "name": "string",  
  "marketModelType": "NOTAPPLICABLE"  
}, {
```

4.2. GET MARKET CALENDAR

This endpoint retrieves calendar information for a specified date and optionally calendar ids..

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date

ids (optional)

Comma-separated list of calendar IDs. format: int64

array[CalendarView]

Content-Type: application/json

```
[
  {
    "calendarId" : 0,
    "dates" : {
      "date" : "date",
      "tradingAllowed" : true,
      {
        "date" : "string" : "date",
        "tradingAllowed" : true
      }
    }
  }, {
    "calendarId" : 0,
    "dates" : [ {
      "date" : "date",
```



```
    "tradingAllowed" : true
  }, {
    "date" : "date",
    "tradingAllowed" : true
  } ]
}
```

4.3. GET SIMPLIFIED LIST OF ALL INSTRUMENTS

```
get
/api/v1/reference/{date}/markets/{mic}/instruments/list
```

This endpoint retrieves a simplified list of all instruments for a specified date, including only key fields.

Path parameters

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date

mic (required)

Market structure's Market Identifier Code.

Return type

array[InstrumentSimplifiedView]

Example data

Content-Type: application/json

```
[ {
  + "code" : "code",
  - "instrumentId" : 0,
  - "code" : "string",
  - "productType" : "FINANCIALPRODUCTSHARE",
  - "productIdentification" : "string" : "productIdentification",
  - "productIdentificationType" : "ISIN",
  - "productType" : "FINANCIALPRODUCTSHARE",
  "productName" : "string" : "productName"
}, {
  "code" : "code",
  "instrumentId" : 0,
  "productIdentification" : "productIdentification",
  "productIdentificationType" : "ISIN",
  "productType" : "FINANCIALPRODUCTSHARE",
  "productName" : "productName"
} ]
```


4.4. GET DETAILED INSTRUMENTS BY MIC

```
get
/api/v1/reference/{date}/markets/{mic}/instruments
```

This endpoint retrieves a detailed list of instruments filtered by a specific MIC for a specified date.

Path parameters

mic (required)

Market structure's Market Identifier Code.

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date

Return type

InstrumentDetailsView

Example data

Content-Type: application/json

```
{
  "tickTables": {
    "key-given_tick_table_id": {
      "tickTableId": 0,
      "ticks": [
        {
          "tickSize": 0,
          "lowerBound": 0
        }
      ]
    }
  },
  "collars": {
    "key-given_collar_group_id": {
      "collarGroupId": 0,
      "orderStaticCollarTableId": 05,
      "orderStaticCollarTable": {
        "collarMode": "TRADEPRICECOLLAR",
        "collarExpression": "PERCENTAGE",
        "collarLowerBound": 0,
        "collarValue": 0,
        "collarLowerBid": 0,
        "collarUpperBid": 0,
        "collarUpperAsk": 0,
        "collarLowerAsk": 0
      },
      "orderDynamicCollarTableId": 07,
      "orderDynamicCollarTable": [
        {
          "collarValue": 7.061401241503109,

```

```

—      "collarMode": "TRADEPRICECOLLAR",
      "collarLowerAsk" : 4.145608029883936,
      "collarUpperBid" : 3.616076749251911,
      "collarLowerBid" : 9.301444243932576,
      "collarUpperAsk" : 2.027123023002322,
      "collarExpression" : "PERCENTAGE",
      "collarLowerBound" : 2.3021358869347655
    }, {
      "collarValue" : 7.061401241503109,
      "collarMode" : "TRADEPRICECOLLAR",
      "collarExpression": "PERCENTAGE", "collarLowerAsk" : 4.145608029883936,
      "collarLowerBound": 0 "collarUpperBid" : 3.616076749251911,
      "collarValue": 0,
—      "collarLowerBid": 0 : 9.301444243932576,
      "collarUpperBid": 0,
—      "collarUpperAsk": 0 : 2.027123023002322,
      "collarLowerAsk": 0 "collarExpression" : "PERCENTAGE",
      "collarLowerBound" : 2.3021358869347655
    } ],
    "tradeStaticCollarTableId": 0, "orderDynamicCollarTable" : [ null, null ],
    "tradeStaticCollarTable": { : [ null, null ],
  {
    "collarMode": "TRADEPRICECOLLAR",
    "collarExpression": "PERCENTAGE",
    "collarLowerBound": 0,
    "collarValue": 0,
    "collarLowerBid": 0,
    "collarUpperBid": 0,
    "collarUpperAsk": 0,
    "collarLowerAsk": 0
  }
},
    "tradeDynamicCollarTableId": 01,
    "tradeStaticCollarTableId" : 1,
    "collarGroupId" : 5,
    "tradeDynamicCollarTable": { : [ null, null ]
  }
},
{
  "collarMode": "TRADEPRICECOLLAR",
  "collarExpression": "PERCENTAGE",
  "collarLowerBound": "instruments" : [ {
    "preTradeCheckMaxQuantity" : 4,
    "instrumentId" : 9,
    "liquidity" : true,
    "nominalValueType" : "NOTAPPLICABLE",
    "settlementCalendarId" : 0,
    "collarValue": "marketModelType" : "NOTAPPLICABLE",
    "collarGroupId" : 5,
    "issuer" : "issuer",
    "productName" : "productName",
    "optionType" : "NOTAPPLICABLE",
    "usIndicator" : "NOTAPPLICABLE",
    "exerciseType" : "AMER",

```

```
"preTradeCheckMaxValue" : 8.969578798196912,  
"productId" : 6,  
"multiplier" : 3.0937452626664474,  
"productIdentification" : "productIdentification",  
"bondCouponType" : "NOTAPPLICABLE",  
"marketStructureId" : 6,  
"tradingScheduleId" : 6,  
"nominalCurrency" : "ALL",  
"strikePrice" : 7.143538047012306,  
"issuerRegCountry" : "AUS",  
"status" : "ACTIVE",  
"code" : "code",  
"mic" : "mic",  
"preTradeCheckMinValue" : 7.740351818741173,  
"priceExpressionType" : "PRICE",  
"expiryDate" : 7,  
"firstTradingDate" : 6,  
"tickTableId" : 6,  
"preTradeCheckMaxPrice" : 6.519180951018382,  
"referencePrice" : 3.2588565619047607,  
"underlyingInstrumentId" : 3,  
"initialPhaseId" : 4,  
"currency" : "ALL",  
"productIdentificationType" : "ISIN",  
"productType" : "FINANCIALPRODUCTSHARE",  
"preTradeCheckMinQuantity" : 0,  
"collarLowerBid" : 7.058770351582356, "settlementType" : "NOTAPPLICABLE",  
"preTradeCheckMinPrice" : 7.058770351582356,  
"lotSize" : 1,  
"cfi" : "cfi",  
"versionNumber" : 5,  
"fisn" : "fisn",  
"productCode" : "productCode",  
"calendarId" : 2,  
"issueSize" : 3,  
"nominalValue" : 3.353193347011243,  
"lastTradingDate" : 3  
}, {  
"preTradeCheckMaxQuantity" : 4,  
"instrumentId" : 9,  
"liquidity" : true,  
"nominalValueType" : "NOTAPPLICABLE",  
"settlementCalendarId" : 0,  
"collarUpperBid" : 7.058770351582356, "marketModelType" : "NOTAPPLICABLE",  
"collarGroupId" : 5,  
"issuer" : "issuer",  
"productName" : "productName",  
"optionType" : "NOTAPPLICABLE",  
"usIndicator" : "NOTAPPLICABLE",  
"exerciseType" : "AMER",  
"preTradeCheckMaxValue" : 8.969578798196912,  
"productId" : 6,  
"multiplier" : 3.0937452626664474,  
"productIdentification" : "productIdentification",
```

```

    "bondCouponType" : "NOTAPPLICABLE",
    "marketStructureId" : 6,
    "tradingScheduleId" : 6,
    "nominalCurrency" : "ALL",
    "strikePrice" : 7.143538047012306,
    "issuerRegCountry" : "AUS",
    "status" : "ACTIVE",
    "code" : "code",
    "mic" : "mic",
    "preTradeCheckMinValue" : 7.740351818741173,
    "priceExpressionType" : "PRICE",
    "expiryDate" : 7,
    "firstTradingDate" : 6,
    "tickTableId" : 6,
    "preTradeCheckMaxPrice" : 6.519180951018382,
    "referencePrice" : 3.2588565619047607,
    "underlyingInstrumentId" : 3,
    "initialPhaseId" : 4,
    "currency" : "ALL",
    "productIdentificationType" : "ISIN",
    "productType" : "FINANCIALPRODUCTSHARE",
    "preTradeCheckMinQuantity" : 0,
    "collarUpperAsk" : 0,    "settlementType" : "NOTAPPLICABLE",
    "preTradeCheckMinPrice" : 7.058770351582356,
    "lotSize" : 1,
    "cfi" : "cfi",
    "versionNumber" : 5,
    "fisn" : "fisn",
    "productCode" : "productCode",
    "calendarId" : 2,
    "issueSize" : 3,
    "nominalValue" : 3.353193347011243,
    "lastTradingDate" : 3
  } ],
  "tickTables" : {
    "key" : {
      "tickTableId" : 0,
    "collarLowerAsk" : 0,    "ticks" : [ {
      "lowerBound" : 1.4658129805029452,
      "tickSize" : 6
    }, {
      "lowerBound" : 1.4658129805029452,
      "tickSize" : 6
    } ]
    }
  },
  "tradingSchedules" : {
    "key" : "given_trading_schedule_id" : {
      "tradingScheduleId" : 0,
      "phases" : [ {
    "tradingPhaseId" : 0,
    "tradingPhaseSettlementCycle" : 0,
    "tradingPhaseMaxBlockSettlementCycle" : 0,

```

```

    "staticCollarVolatilityAuctionId": 0,
    "dynamicCollarVolatilityAuctionId": 5,
    "tradingPhaseStartTime": 9,
    "tradingPhaseType": "CONTINUOUSPRICETIME",
    "auctionType": "NOTAPPLICABLE",
    "uncrossing": true,
    "extStaticCollarVolatilityAuctionId": 9,
    "extDynamicCollarVolatilityAuctionId": 6,
    "tradingPhaseType" : "CONTINUOUSPRICETIME",
    "tradingPhaseSettlementCycle" : 7,
    "auctionType" : "NOTAPPLICABLE",
    "staticCollarVolatilityAuctionId" : 4,
    "uncrossing" : true,
    "lastAuctionPhaseId": 8,
    "tradingPhaseMaxBlockSettlementCycle" : 1
}, {
}
    "instruments": {
    {
        "instrumentId": 0,
        "productType": "FINANCIALPRODUCTSHARE",
        "firstTradingDate": 0,
        "lastTradingDate": 0,
        "productId": 0,
        "currency": "ALL",
        "lotSize": 0,
        "priceExpressionType": "PRICE",
        "calendarId": 0,
        "marketStructureId": 0,
        "tickTableId": 0,
        "collarGroupId": 0,
        "tradingScheduleId": 0,
        "code": "string",
        "mie": "string",
        "nominalValue": 0,
        "nominalValueType": "tradingPhaseId" : 6,
        "dynamicCollarVolatilityAuctionId" : 5,
        "tradingPhaseStartTime" : 9,
        "extStaticCollarVolatilityAuctionId" : 9,
        "extDynamicCollarVolatilityAuctionId" : 6,
        "tradingPhaseType" : "CONTINUOUSPRICETIME",
        "tradingPhaseSettlementCycle" : 7,
        "auctionType" : "NOTAPPLICABLE",
        "multiplier": 0,
        "strikePrice": 0,
        "productIdentification": "string",
        "productIdentificationType": "ISIN",
        "settlementCalendarId": 0,
        "bondCouponType": "NOTAPPLICABLE",
        "preTradeCheckMinPrice": 0,
        "preTradeCheckMaxPrice": 0,
        "preTradeCheckMinQuantity": 0,
        "preTradeCheckMaxQuantity": 0,

```

```

"preTradeCheckMaxValue": 0,
"preTradeCheckMinValue": 0,
"liquidity": "staticCollarVolatilityAuctionId" : 4,
"uncrossing" : true,
"marketModelType": "NOTAPPLICABLE", "lastAuctionPhaseId" : 8,
"issuer": "string",
"issuerRegCountry": "AUS",
"underlyingInstrumentId": 0,
"productCode": "string",
"efi": "string",
"fion": "string",
"issueSize": 0,
"nominalCurrency": "ALL",
"usIndicator": "NOTAPPLICABLE",
"expiryDate": 0,
"versionNumber": 0,
"settlementType": "NOTAPPLICABLE",
"optionType": "NOTAPPLICABLE",
"exerciseType": "AMER",
"productName": "string",
"referencePrice": 0,
"status": "ACTIVE",
"initialPhaseId": 0
}
}
"tradingPhaseMaxBlockSettlementCycle" : 1
}
}
}
}

```

4.5. GET DETAILED INSTRUMENTS BY IDS

```
get /api/v1/reference/{date}/instruments
```

This endpoint retrieves detailed information about specified instruments for a given date, filtered by their IDs.

Path parameters

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date

Query parameters

ids (required)

Comma-separated list of instrument IDs. format: int64

Return type

InstrumentDetailView

Example data

Content-Type: application/json

```

{
  "tickTables": {
  "key-given-tick-table-id": {
    "tickTableId": 0,
    "ticks": {
      {
        "tickSize": 0,
        "lowerBound": 0
      }
    }
  },
  "collars":": {
    "key-given-collar-group-id":": {
      "collarGroupId": 0,
      "orderStaticCollarTableId":: 05,
      "orderStaticCollarTable": {
        {
          "collarMode": "TRADEPRICECOLLAR",
          "collarExpression": "PERCENTAGE",
          "collarLowerBound": 0,
          "collarValue": 0,
          "collarLowerBid": 0,
          "collarUpperBid": 0,
          "collarUpperAsk": 0,
          "collarLowerAsk": 0
        }
      }
    },
    "orderDynamicCollarTableId":: 07,
    "orderDynStaticCollarTable":: [ {
      +"collarValue" : 7.061401241503109,
      "collarMode":" : "TRADEPRICECOLLAR",
      "collarLowerAsk" : 4.145608029883936,
      "collarUpperBid" : 3.616076749251911,
      "collarLowerBid" : 9.301444243932576,
      "collarUpperAsk" : 2.027123023002322,
      "collarExpression" : "PERCENTAGE",
      "collarLowerBound" : 2.3021358869347655
    }, {
      "collarValue" : 7.061401241503109,
      "collarMode" : "TRADEPRICECOLLAR",
      "collarExpression": "PERCENTAGE","collarLowerAsk" : 4.145608029883936,
      "collarLowerBound": 0"collarUpperBid" : 3.616076749251911,
      "collarValue": 0,
      "collarLowerBid": 0 : 9.301444243932576,
      "collarUpperBid": 0,
      "collarUpperAsk": 0 : 2.027123023002322,
      "collarLowerAsk": 0"collarExpression" : "PERCENTAGE",
      "collarLowerBound" : 2.3021358869347655
    }
  ],
  "tradeStaticCollarTableId": 0,orderDynamicCollarTable" : [ null, null ],
  "tradeStaticCollarTable": {" : [ null, null ],

```



```

    {
        "collarMode": "TRADEPRICECOLLAR",
        "collarExpression": "PERCENTAGE",
        "collarLowerBound": 0,
        "collarValue": 0,
        "collarLowerBid": 0,
        "collarUpperBid": 0,
        "collarUpperAsk": 0,
        "collarLowerAsk": 0
    },
},
    "tradeDynamicCollarTableId": 01,
    "tradeStaticCollarTableId" : 1,
    "collarGroupId" : 5,
    "tradeDynamicCollarTable": { : [ null, null ]
}
},
    {
        "collarMode": "TRADEPRICECOLLAR",
        "collarExpression": "PERCENTAGE",
        "collarLowerBound": "instruments" : [ {
            "preTradeCheckMaxQuantity" : 4,
            "instrumentId" : 9,
            "liquidity" : true,
            "nominalValueType" : "NOTAPPLICABLE",
            "settlementCalendarId" : 0,
        "collarValue": "marketModelType" : "NOTAPPLICABLE",
            "collarGroupId" : 5,
            "issuer" : "issuer",
            "productName" : "productName",
            "optionType" : "NOTAPPLICABLE",
            "usIndicator" : "NOTAPPLICABLE",
            "exerciseType" : "AMER",
            "preTradeCheckMaxValue" : 8.969578798196912,
            "productId" : 6,
            "multiplier" : 3.0937452626664474,
            "productIdentification" : "productIdentification",
            "bondCouponType" : "NOTAPPLICABLE",
            "marketStructureId" : 6,
            "tradingScheduleId" : 6,
            "nominalCurrency" : "ALL",
            "strikePrice" : 7.143538047012306,
            "issuerRegCountry" : "AUS",
            "status" : "ACTIVE",
            "code" : "code",
            "mic" : "mic",
            "preTradeCheckMinValue" : 7.740351818741173,
            "priceExpressionType" : "PRICE",
            "expiryDate" : 7,
            "firstTradingDate" : 6,
            "tickTableId" : 6,
            "preTradeCheckMaxPrice" : 6.519180951018382,
            "referencePrice" : 3.2588565619047607,
            "underlyingInstrumentId" : 3,

```

```
"initialPhaseId" : 4,  
"currency" : "ALL",  
"productIdentificationType" : "ISIN",  
"productType" : "FINANCIALPRODUCTSHARE",  
"preTradeCheckMinQuantity" : 0,  
"collarLowerBid" : "settlementType" : "NOTAPPLICABLE",  
"preTradeCheckMinPrice" : 7.058770351582356,  
"lotSize" : 1,  
"cfi" : "cfi",  
"versionNumber" : 5,  
"fisn" : "fisn",  
"productCode" : "productCode",  
"calendarId" : 2,  
"issueSize" : 3,  
"nominalValue" : 3.353193347011243,  
"lastTradingDate" : 3  
}, {  
  "preTradeCheckMaxQuantity" : 4,  
  "instrumentId" : 9,  
  "liquidity" : true,  
  "nominalValueType" : "NOTAPPLICABLE",  
  "settlementCalendarId" : 0,  
  "collarUpperBid" : "marketModelType" : "NOTAPPLICABLE",  
  "collarGroupId" : 5,  
  "issuer" : "issuer",  
  "productName" : "productName",  
  "optionType" : "NOTAPPLICABLE",  
  "usIndicator" : "NOTAPPLICABLE",  
  "exercerciseType" : "AMER",  
  "preTradeCheckMaxValue" : 8.969578798196912,  
  "productId" : 6,  
  "multiplier" : 3.0937452626664474,  
  "productIdentification" : "productIdentification",  
  "bondCouponType" : "NOTAPPLICABLE",  
  "marketStructureId" : 6,  
  "tradingScheduleId" : 6,  
  "nominalCurrency" : "ALL",  
  "strikePrice" : 7.143538047012306,  
  "issuerRegCountry" : "AUS",  
  "status" : "ACTIVE",  
  "code" : "code",  
  "mic" : "mic",  
  "preTradeCheckMinValue" : 7.740351818741173,  
  "priceExpressionType" : "PRICE",  
  "expiryDate" : 7,  
  "firstTradingDate" : 6,  
  "tickTableId" : 6,  
  "preTradeCheckMaxPrice" : 6.519180951018382,  
  "referencePrice" : 3.2588565619047607,  
  "underlyingInstrumentId" : 3,  
  "initialPhaseId" : 4,  
  "currency" : "ALL",  
  "productIdentificationType" : "ISIN",  
  "productType" : "FINANCIALPRODUCTSHARE",
```

```

    "preTradeCheckMinQuantity" : 0,
    "collarUpperAsk" :     "settlementType" : "NOTAPPLICABLE",
    "preTradeCheckMinPrice" : 7.058770351582356,
    "lotSize" : 1,
    "cfi" : "cfi",
    "versionNumber" : 5,
    "fisn" : "fisn",
    "productCode" : "productCode",
    "calendarId" : 2,
    "issueSize" : 3,
    "nominalValue" : 3.353193347011243,
    "lastTradingDate" : 3
  } ],
  "tickTables" : {
    "key" : {
      "tickTableId" : 0,
      "collarLowerAsk" : 0      "ticks" : [ {
        "lowerBound" : 1.4658129805029452,
        "tickSize" : 6
      }, {
        "lowerBound" : 1.4658129805029452,
        "tickSize" : 6
      } ]
    }
  },
  "tradingSchedules" : {
    "key" :     "key-given_trading_schedule_id" : {
      "tradingScheduleId" : 1,
      "phases" : [ {
        {
          "tradingPhaseId" : 6,
          "tradingPhaseSettlementCycle" : 0,
          "tradingPhaseMaxBlockSettlementCycle" : 0,
          "staticCollarVolatilityAuctionId" : 0,
          "dynamicCollarVolatilityAuctionId" : 5,
          "tradingPhaseStartTime" : 9,
          "tradingPhaseType" : "CONTINUOUSPRICETIME",
          "auctionType" : "NOTAPPLICABLE",
          "uncrossing" : true,
          "extStaticCollarVolatilityAuctionId" : 9,
          "extDynamicCollarVolatilityAuctionId" : 6,
          "tradingPhaseType" : "CONTINUOUSPRICETIME",
          "tradingPhaseSettlementCycle" : 7,
          "auctionType" : "NOTAPPLICABLE",
          "staticCollarVolatilityAuctionId" : 4,
          "uncrossing" : true,
          "lastAuctionPhaseId" : 8,
          "tradingPhaseMaxBlockSettlementCycle" : 1
        }, {
      } ]
    }
  }
  "instruments" : {
  {
    "instrumentId" : 0,

```

```

"productType": "FINANCIALPRODUCTSHARE",
"firstTradingDate": 0,
"lastTradingDate": 0,
"productId": 0,
"currency": "ALL",
"lotSize": 0,
"priceExpressionType": "PRICE",
"calendarId": 0,
"marketStructureId": 0,
"tickTableId": 0,
"collarGroupId": 0,
"tradingScheduleId": 0,
"code": "string",
"mic": "string",
"nominalValue": 0,
"nominalValueType": "tradingPhaseId" : 6,
"dynamicCollarVolatilityAuctionId" : 5,
"tradingPhaseStartTime" : 9,
"extStaticCollarVolatilityAuctionId" : 9,
"extDynamicCollarVolatilityAuctionId" : 6,
"tradingPhaseType" : "CONTINUOUSPRICETIME",
"tradingPhaseSettlementCycle" : 7,
"auctionType" : "NOTAPPLICABLE",
"multiplier": 0,
"strikePrice": 0,
"productIdentification": "string",
"productIdentificationType": "ISIN",
"settlementCalendarId": 0,
"bondCouponType": "NOTAPPLICABLE",
"preTradeCheckMinPrice": 0,
"preTradeCheckMaxPrice": 0,
"preTradeCheckMinQuantity": 0,
"preTradeCheckMaxQuantity": 0,
"preTradeCheckMaxValue": 0,
"preTradeCheckMinValue": 0,
"liquidity": "staticCollarVolatilityAuctionId" : 4,
"uncrossing" : true,
"marketModelType": "NOTAPPLICABLE", "lastAuctionPhaseId" : 8,
"issuer": "string",
"issuerRegCountry": "AUS",
"underlyingInstrumentId": 0,
"productCode": "string",
"efi": "string",
"fisn": "string",
"issueSize": 0,
"nominalCurrency": "ALL",
"usIndicator": "NOTAPPLICABLE",
"expiryDate": 0,
"versionNumber": 0,
"settlementType": "NOTAPPLICABLE",
"optionType": "NOTAPPLICABLE",
"exerciseType": "AMER",
"productName": "string",
"referencePrice": 0,

```

```
"status": "ACTIVE",  
"initialPhaseId": 0  
}  
}  
    "tradingPhaseMaxBlockSettlementCycle" : 1  
  } 1  
}  
}
```

4.6. GET INDEXES METADATA AND PORTFOLIO

```
get /api/v1/reference/{date}/indexes
```

This endpoint retrieves Index metadata and portfolio on a specific date.

Path parameters

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date

Query parameters

ids (optional)

Comma-separated list of instrument Ids. format: int64

Return type

array[IndexReferenceDataView]

Example data

Content-Type: application/json

```
[ {  
  "productName": "string",  
  "portfolio" : [ {  
    "instrumentPacket" : 4,  
    "mic" : "mic",  
    "currency" : "ALL"  
  }, {  
    "instrumentPacket" : 4,  
    "mic" : "mic",  
    "currency" : "ALL"  
  } ],  
  "params" : {  
    "numberOfDividends" : 9,  
    "correctionCoefficient" : 5.637376656633329,  
    "indexUnderlying" : [ {  
      "productIdentification" : "productIdentification",  
      "productIdentificationType" : "ISIN"  
    }, {  

```

```

    "productIdentification" : "productIdentification",
    "productIdentificationType" : "ISIN"
  } ],
  "instrumentId" : 0,
  "indexPublicationSchedule" : "CONTINUOUS",
  "indexType" : "W",
  "numOfComponents" : 2,
  "daysSinceLastPublication" : 7,
  "dateValidity" : 2,
  "indexBaseValue" : 6.027456183070403,
  "currency" : "ALL",
  "indexBaseCapitalisation" : 1.4658129805029452,
  "indexBaseDate" : 05,
  "correctionCoefficient" : 0,
  "numOfComponents" : 0,
  "indexPublicationSchedule" : "CONTINUOUS",
  "indexType" : "W",
  "daysSinceLastPublication" : 0,
  "numberOfDividends" : 0,
  "indexUnderlying" : {
    {
      "productIdentification" : "string",
      "productIdentificationType" : "ISIN"
    }
  },
  "publicationOrder" : 03,
  "currency" : "ALL",
  "dateValidity" : 0
},
{
  "productName" : "productName"
}, {
  "portfolio" : [ {
    { "instrumentPacket" : 4,
      "publicProductIdentification" : {
        "mic" : "mic",
        "currency" : "ALL"
      }
    }, {
      "instrumentPacket" : 4,
      "mic" : "mic",
      "currency" : "ALL"
    }
  ] },
  "params" : {
    "numberOfDividends" : 9,
    "correctionCoefficient" : 5.637376656633329,
    "indexUnderlying" : [ {
      "productIdentification" : "productIdentification",
      "productIdentificationType" : "ISIN"
    } ],
    {
      "mic" : "string",
      "productIdentification" : "productIdentification",
      "productIdentificationType" : "ISIN"
    }
  ] },
  "instrumentId" : 0,

```

```

    "indexPublicationSchedule" : "CONTINUOUS",
    "indexType" : "W",
    "numOfComponents" : 2,
    "daysSinceLastPublication" : 7,
    "dateValidity" : 2,
    "indexBaseValue" : 6.027456183070403,
    "currency": "ALL",
    "instrumentPacket": 0
  }
}
}
}
    "indexBaseCapitalisation" : 1.4658129805029452,
    "indexBaseDate" : 5,
    "publicationOrder" : 3
  },
  "productName" : "productName"
} ]

```

4.7. GET ACCRUED INTEREST TABLE

```

get
/api/v1/reference/{date}/markets/{mic}/accrued-
interest-tables

```

This endpoint retrieves accrued interest data for financial instruments associated with a specific market on a given date.

Path parameters

mic (required)

Market structure's Market Identifier Code.

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date

Return type

array[AccruedInterestTableView]

Example data

Content-Type: application/json

```

[ {
  "productId" : 1,
  "productIdentification" : "productIdentification",
  "accruedInterestTable": [ {
    "date": 0,
    "value": 6.027456183070403
  }, {
    "date" : 0,
    "value" : 6.027456183070403
  } ]
} ]

```

```

"productId": 0,
"productIdentification": "string",
"productIdentificationType": "ISIN"
}
}, {
  "productId" : 1,
  "productIdentification" : "productIdentification",
  "accruedInterestTable" : [ {
    "date" : 0,
    "value" : 6.027456183070403
  }, {
    "date" : 0,
    "value" : 6.027456183070403
  } ],
  "productIdentificationType" : "ISIN"
}]

```

4.8. GET KEY INFORMATION DATA FOR INSTRUMENTS

```
get /api/v1/reference/{date}/markets/{mic}/kids
```

This endpoint retrieves Key Information Documents (KIDs) for instruments in a specific market on a given date.

Path parameters

mic (required)

Market structure's Market Identifier Code.

date (required)

The date for which data is requested, in the format YYYY-MM-DD. format: date

Return type

array[InstrumentKidView]

Example data

Content-Type: application/json

```

[ {
  {
"productId": "string", : 0,
"productIdentification": "string" : "productIdentification",
"productIdentificationType": "ISIN",
"url": "string" : "url"
}
}, {
  "productId" : 0,
  "productIdentification" : "productIdentification",
  "productIdentificationType" : "ISIN",
  "url" : "url"
}]

```


5. MARKET DATA ENDPOINTS

5.1. GET INSTRUMENT SNAPSHOT INFORMATION

```
get /api/v1/snapshot/instrument/{id}
```

This endpoint retrieves information about a specific financial instrument, including changes to instruments, instrument information, book information, trade information, and collars for the given moment. The 'id' parameter specifies the instrument.

Path parameters

id (required)

The unique identifier for the financial instrument. format: int64

Return type

InstrumentSnapshotView

Example data

Content-Type: application/json.

```
{
  "timestamp": "string", tradingPhaseId : 7,
  "referencePrice": 0,
  "totalVolume" : 8.762042012749001,
  "totalValue" : 9.018348186070783,
  "pctChange" : 4.965218492984954,
  "code" : "code",
  "priceLevelSnapshot" : {
    "maxDepth" : 9,
    "buy" : [ {
      "quantity" : 2,
      "price" : 3.616076749251911,
      "orderCount" : 4
    }, {
      "quantity" : 2,
      "price" : 3.616076749251911,
      "orderCount" : 4
    } ],
    "sell" : [ null, null ]
  },
  "instrumentId": 46,
  "code": "string",
  "vwap" : 6.683562403749608,
  "productIdentification": "string" : "productIdentification",
  "productIdentificationType": "ISIN",
  "openPrice" : 5.025004791520295,
  "instrumentStatus" : "ACTIVE",
  "noTrades" : 6,
  "referencePrice" : 0.8008281904610115,
```

```
|  |
| --- |
| "tradingPhaseType" : "CONTINUOUSPRICETIME", |
| "staticCollars" : { |
| "collarsReferencePrice" : 7.386281948385884, |
| "upperAsk" : 1.1730742509559433, |
| "lowerTrade" : 1.2315135367772556, |
| "upperTrade" : 1.0246457001441578, |
| "lowerBid" : 1.4894159098541704, |
| "upperBid" : 6.84685269835264, |
| "lowerAsk" : 7.457744773683766 |
| }, |
| "minPrice" : 9.369310271410669, |
| "auctionUpdate"+ : { |
| "indicativeMatchingPrice": 10.1totalSellQuantityAtImp" : 5, |
| "bestSellLevelQuantity" : 2, |
| "indicativeMatchingVolume" : 01, |
| "totalSellQuantityAtImp": 0bestBuyLevel" : 10.1, |
| "bestSellLevel" : 10.1, |
| "totalBuyQuantityAtImp" : 05, |
| "bestSellLevel": 10.1, |
| "bestSellLevelQuantity": 0, |
| "bestBuyLevel": 10.1, |
| "bestBuyLevelQuantity"+ : 123456, |
| "auctionType"+ : "NOTAPPLICABLE"+, |
| }, |
| "instrumentStatus": "ACTIVE", |
| "tradingPhaseId": 0, |
| "tradingPhaseType": "CONTINUOUSPRICETIME", |
| "priceLevelSnapshot": { |
| "maxDepth": 0, |
| "buy": { |
| { |
| "price": 0, |
| "quantity": 0, |
| "orderCount": 0 |
| } |
| } |
| "sell": { |
| { |
| "price": 0, |
| "quantity": 0, |
| "orderCount": 0 |
| } |
| } |
| }, |
| "staticCollars": { |
| "collarsReferencePrice": 0, |
| "lowerTrade": 0, |
| "upperTrade": 0, |
| "lowerBid": 0, |
| "upperBid": 0, |
| "lowerAsk": 0, |
| "upperAsk": 0 |
| }, |
| "dynamicCollars": { |

```

```

"collarsReferencePrice": 0,
"lowerTrade": 0,
"upperTrade": 0,
"lowerBid": 0,
"upperBid": 0,
"lowerAsk": 0,
"upperAsk": 0
},
"pctChange": 0,
"openPrice": 0,
  "indicativeMatchingPrice" : 10.1
},
  "maxPrice": 9.965781217890562,
"minPrice": 0,
"vwap": 0,
"totalVolume": 0,
"totalValue": 0,
"noTrades": 0
  "productIdentificationType" : "ISIN",
  "timestamp" : "timestamp"
}

```

5.2. GET INDEX SNAPSHOT DATA

GET /api/v1/snapshot/index/{id}

This endpoint retrieves snapshot data for a specified index, including metrics like index level code, number of active instruments, session highs/lows, and bid/ask values.

Path parameters

id (required)

Unique identifier for the index for which real-time data is requested. format: int64

Return type

RealTimeIndexView

Example data

Content-Type: application/json

```

{
"instrumentId": 0,
"indexLevelCode": "PREVIOUSCLOSINGVALUE",
"numOfActiveInstruments": 0,
  "indIndexOpeningPortfolio": 1.4658129805029452,
  "indexValue": 6,
  "instrumentId" : 0,
  "tradingValue" : 2.3021358869347655,
  "pctChangeIndexValPrevSession": 5.637376656633329,
"tradingValue": 0,
  "sessionLow": 7.061401241503109,
"sessionHigh": 0,

```

```
"indexValueBid": 0,  
"indexValueAsk": 0,  
  "midSpreadIndex" : 4.145608029883936,  
  "differenceCentralSpread" : 7.386281948385884,  
  "indexValueAsk" : 2.027123023002322,  
  "sessionHigh" : 9.301444243932576,  
  "indexLevelCode" : "PREVIOUSCLOSINGVALUE",  
  "indexValueBid" : 3.616076749251911,  
  "indexValue" : 5.962133916683182  
}
```

5.3. GET INSTRUMENT SUMMARY LIST

GET /api/v1/summary/instrument

This endpoint retrieves a list of instrument summaries.

Query parameters

ids (required)

Comma-separated list of instrument IDs. format: int64

Return type

array[InstrumentSummaryView]

Example data

Content-Type: application/json

```
[  
  {  
"lastTradedPrice": 0,  
"closingPrice": 0,  
    "totalVolume" : 7,  
    "totalValue" : 9.301444243932576,  
    "closingPriceType" : "LTP",  
"pctChange" : 5.637376656633329,  
    "instrumentId" : 0,  
    "adjustedClosingPrice" : 5.962133916683182,  
"vwap" : 7.386281948385884,  
    "adjustedClosingPriceReason" : "REGULAR",  
"pctChange": 0,  
"noTrades": 0,  
"totalVolume": 0,  
"totalValue": 0,  
"openingPrice": 0 : 3.616076749251911,  
    "noTrades" : 2,  
    "minPrice" : 4.145608029883936,  
    "closingPrice" : 1.4658129805029452,  
    "maxPrice" : 2.027123023002322,  
"minPrice": "lastTradedPrice" : 6.027456183070403  
  },  
  {  
    "totalVolume" : 7,  

```

```

    "totalValue" : 9.301444243932576,
    "closingPriceType" : "LTP",
    "pctChange" : 5.637376656633329,
    "instrumentId" : 0,
    "adjustedClosingPrice" : 5.962133916683182,
    "vwap" : 7.386281948385884,
    "adjustedClosingPriceReason" : "REGULAR",
    "openingPrice" : 3.616076749251911,
    "noTrades" : 2,
    "minPrice" : 4.145608029883936,
    "closingPrice" : 1.4658129805029452,
    "maxPrice" : 2.027123023002322,
    "lastTradedPrice" : 6.027456183070403
  } ]

```

5.4. GET INDEX SUMMARY LIST

```
get /api/v1/summary/index
```

This endpoint retrieves a list of index summaries, providing key index metrics such as opening value, session highs/lows, and closing values.

Query parameters

ids (required)

Comma-separated list of instrument IDs. format: int64

Return type

array[InstrumentSummaryView]

Example data

Content-Type: application/json

```

[ {
  "sessionAvg" : 2.3021358869347655,
  "pctChangeIndexValPrevSession" : 7.061401241503109,
  "indIndexOpeningPortfolio" : 2.027123023002322,
  "sessionLow" : 1.4658129805029452,
  "closingCapitalisationPortfolio" : 3.616076749251911,
  "instrumentId" : 0,
  "openingIndexValue" : 6.027456183070403,
  "sessionLow" : 0,
  "sessionHigh" : 5.962133916683182,
  "closingIndexValue" : 5.637376656633329,
  "sessionAvg" : 0,
  "pctChangeIndexValPrevSession" : 0,
  "pctChangeIndexValPrevYear" : 9.301444243932576
}, {
  "sessionAvg" : 2.3021358869347655,
  "pctChangeIndexValPrevSession" : 7.061401241503109,

```

```
"indIndexOpeningPortfolio" : 2.027123023002322,  
"sessionLow" : 1.4658129805029452,  
"closingCapitalisationPortfolio" : 3.616076749251911,  
"instrumentId" : 0,  
"indIndexOpeningPortfolio": 0  
}  
"openingIndexValue" : 6.027456183070403,  
"sessionHigh" : 5.962133916683182,  
"closingIndexValue" : 5.637376656633329,  
"pctChangeIndexValPrevYear" : 9.301444243932576  
}_]
```

6. SCHEMAFORKAFKA ENDPOINTS

6.1. GET LATEST AVRO SCHEMA FOR KAFKA STREAM

```
get /api/v1/schema/avro
```

Return type

String

Example data

Content-Type: application/json

```
avro schema
```

7. MODELS

7.1. LIST OF MODELS

1. AccruedInterestEntryView
2. AccruedInterestTableView
3. AuctionUpdateView
4. CalendarDayView
5. CalendarView
6. CollarGroupView
7. CollarTableEntryView
8. CollarsView
9. IndexParamsView
10. IndexPortfolioEntryView
11. IndexReferenceDataView
12. IndexSummaryView
13. InstrumentDetailsView
14. InstrumentKidView
15. InstrumentSimplifiedView
16. InstrumentSnapshotView
17. InstrumentSummaryView
18. InstrumentView
19. MarketStructureView
20. PriceLevelSnapshotView
21. PriceLevelView
22. PublicProductIdentificationView
23. RealTimeIndexView
24. TickTableView
25. TickView
26. TradingPhaseView
27. TradingScheduleView

7.1.1. ACCRUEDINTERESTENTRYVIEW

Message describing an accrued interest table entry.

date (optional)

Long Accrued interest table entry date. format: int64

value (optional)

BigDecimal Accrued interest table entry value.

7.1.2. ACCRUEDINTERESTTABLEVIEW

Message describing an accrued interest table entry for product.

accruedInterestTable (optional)

array[AccruedInterestEntryView] Accrued interest table.

productId (optional)

Long Product ID. format: int64

productIdentification (optional)

String Product identification, e.g. ISIN number.

productIdentificationType (optional)

String Type of product identification (for example ISIN or SEDOL).

Enum:

ISIN

~~UNKNOWN~~

7.1.3. AUCTIONUPDATEVIEW

Updates IMP/IMV and BBO during an auction.

indicativeMatchingPrice (optional)

BigDecimal Indicative Matching Price (IMP).

example: 10.1

indicativeMatchingVolume (optional)

Integer Indicative Matching Volume (IMV).

totalSellQuantityAtImp (optional)

Integer Total sell quantity at IMP.

totalBuyQuantityAtImp (optional)

Integer Total buy quantity at IMP.

bestSellLevel (optional)

BigDecimal Best sell price level (1st BBO level).

example: 10.1

bestSellLevelQuantity (optional)

Integer Quantity at the best sell price level.

bestBuyLevel (optional)

BigDecimal Best buy price level (1st BBO level).

example: 10.1

bestBuyLevelQuantity (optional)

Integer Quantity at the best buy price level.

example: 123456

auctionType (optional)

String Type of auction

Enum:

NOTAPPLICABLE

OPENINGAUCTION

CLOSINGAUCTION

INTRADAYAUCATION

VOLATILITYAUCTIONSTATIC

VOLATILITYAUCTIONDYNAMIC

EXTENDEDVOLATILITYAUCTIONSTATIC

EXTENDEDVOLATILITYAUCTIONDYNAMIC

UNSUSPENSIONAUCTION

~~UNKNOWN~~

7.1.4. CALENDARDAYVIEW

tradingAllowed (optional)

Boolean Is trading allowed parameter for a given day

date (optional)

String Date in format YYYY:mm:dd

7.1.5. CALENDARVIEW

Calendar for a given instrument for a given day

dates (optional)

array[CalendarDayView]

calendarId (optional)

Long ID of the calendar format: int64

7.1.6. COLLARGROUPVIEW

Collar group.

collarGroupId (optional)

Long Collar group ID. format: int64

orderStaticCollarTableId (optional)

Long Order static collar table ID. format: int64

orderStaticCollarTable (optional)

array[CollarTableEntryView] Order static collars.

orderDynamicCollarTableId (optional)

Long Order dynamic collar table ID. format: int64

orderDynamicCollarTable (optional)

array[CollarTableEntryView] Order dynamic collars.

tradeStaticCollarTableId (optional)

Long Trade static collar table ID. format: int64

tradeStaticCollarTable (optional)

array[CollarTableEntryView] Trade static collars.

tradeDynamicCollarTableId (optional)

Long Trade dynamic collar table ID. format: int64

tradeDynamicCollarTable (optional)

array[CollarTableEntryView] Trade dynamic collars.

7.1.7. COLLARTABLEENTRYVIEW

Collar table definition

collarMode (optional)

String Collar mode (trade price collar or order price collar)

Enum:

TRADEPRICECOLLAR

ORDERPRICECOLLAR

~~UNKNOWN~~

collarExpression (optional)

String Expression Type

Enum:

PERCENTAGE

ABSOLUTEVALUE

~~UNKNOWN~~

collarLowerBound (optional)

BigDecimal Collar lower bound.

collarValue (optional)

BigDecimal Collar value.

collarLowerBid (optional)

BigDecimal Lower bid collar.

collarUpperBid (optional)

BigDecimal Upper bid collar.

collarUpperAsk (optional)

BigDecimal Upper ask collar.

collarLowerAsk (optional)

BigDecimal Lower ask collar.

7.1.8. COLLARSVIEW

New order price collars

collarsReferencePrice (optional)

BigDecimal The reference price.

lowerTrade (optional)

BigDecimal Value of the lower bid Trade collar.

upperTrade (optional)

BigDecimal Value of the upper bid Trade collar.

lowerBid (optional)

BigDecimal Value of the lower bid Price collar.

upperBid (optional)

BigDecimal Value of the upper bid Price collar.

lowerAsk (optional)

BigDecimal Value of the lower ask Price collar.

upperAsk (optional)

BigDecimal Value of the upper ask Price collar.

7.1.9. INDEXPARAMSVIEW

Index parameters metadata

instrumentId (optional)

Long ID of financial instrument. format: int64

indexBaseValue (optional)

BigDecimal Index base value.

indexBaseCapitalisation (optional)

BigDecimal Index base capitalization.

indexBaseDate (optional)

Long Index base date. format: int64

correctionCoefficient (optional)

BigDecimal Index adjustment coefficient K(t).

numOfComponents (optional)

Integer Total number of instruments in the portfolio. format: int32

indexPublicationSchedule (optional)

String Mode of index dissemination

Enum:

CONTINUOUS

FIXING

CLOSING

~~UNKNOWN~~

indexType (optional)

String Type of index

Enum:

W

Z

V

M

S

L

~~UNKNOWN~~

daysSinceLastPublication (optional)

Integer Number of calendar days between the current session and the previous session day. format: int32

numberOfDividends (optional)

Integer Number of dividends included in the index calculation. format: int32

indexUnderlying (optional)

array[PublicProductIdentificationView] List of underlying instruments for the index.

publicationOrder (optional)

Integer Index publication presentation order. format: int32

currency (optional)

String Currency - ISO 4217 (2022/04/01) code (e.g. PLN)

Enum:

ALL - Lek

DZD - Algerian Dinar

ARS - Argentine Peso

AUD - Australian Dollar

BSD - Bahamian Dollar

BHD - Bahraini Dinar

BDT - Taka
AMD - Armenian Dram
BBD - Barbados Dollar
BMD - Bermudian Dollar
BTN - Ngultrum
BOB - Boliviano
BWP - Pula
BZD - Belize Dollar
SBD - Solomon Islands Dollar
BND - Brunei Dollar
MMK - Kyat
BIF - Burundi Franc
KHR - Riel
CAD - Canadian Dollar
CVE - Cabo Verde Escudo
KYD - Cayman Islands Dollar
LKR - Sri Lanka Rupee
CLP - Chilean Peso
CNY - Yuan Renminbi
COP - Colombian Peso
KMF - Comorian Franc
CRC - Costa Rican Colon
HRK - Kuna
CUP - Cuban Peso
CZK - Czech Koruna
DKK - Danish Krone
DOP - Dominican Peso
SVC - El Salvador Colon
ETB - Ethiopian Birr
ERN - Nakfa
FKP - Falkland Islands Pound
FJD - Fiji Dollar
DJF - Djibouti Franc
GMD - Dalasi
GIP - Gibraltar Pound
GTQ - Quetzal
GNF - Guinean Franc
GYD - Guyana Dollar
HTG - Gourde
HNL - Lempira
HKD - Hong Kong Dollar
HUF - Forint
ISK - Iceland Krona

INR - Indian Rupee
IDR - Rupiah
IRR - Iranian Rial
IQD - Iraqi Dinar
ILS - New Israeli Sheqel
JMD - Jamaican Dollar
JPY - Yen
KZT - Tenge
JOD - Jordanian Dinar
KES - Kenyan Shilling
KPW - North Korean Won
KRW - Won
KWD - Kuwaiti Dinar
KGS - Som
LAK - Lao Kip
LBP - Lebanese Pound
LSL - Loti
LRD - Liberian Dollar
LYD - Libyan Dinar
MOP - Pataca
MWK - Malawi Kwacha
MYR - Malaysian Ringgit
MVR - Rufiyaa
MUR - Mauritius Rupee
MXN - Mexican Peso
MNT - Tugrik
MDL - Moldovan Leu
MAD - Moroccan Dirham
OMR - Rial Omani
NAD - Namibia Dollar
NPR - Nepalese Rupee
ANG - Netherlands Antillean Guilder
AWG - Aruban Florin
VUV - Vatu
NZD - New Zealand Dollar
NIO - Cordoba Oro
NGN - Naira
NOK - Norwegian Krone
PKR - Pakistan Rupee
PAB - Balboa
PGK - Kina
PYG - Guarani
PEN - Sol

PHP - Philippine Peso
QAR - Qatari Rial
RUB - Russian Ruble
RWF - Rwanda Franc
SHP - Saint Helena Pound
SAR - Saudi Riyal
SCR - Seychelles Rupee
SGD - Singapore Dollar
VND - Dong
SOS - Somali Shilling
ZAR - Rand
SSP - South Sudanese Pound
SZL - Lilangeni
SEK - Swedish Krona
CHF - Swiss Franc
SYP - Syrian Pound
THB - Baht
TOP - Pa'anga
TTD - Trinidad and Tobago Dollar
AED - UAE Dirham
TND - Tunisian Dinar
UGX - Uganda Shilling
MKD - Denar
EGP - Egyptian Pound
GBP - Pound Sterling
TZS - Tanzanian Shilling
USD - US Dollar
UYU - Peso Uruguayo
UZS - Uzbekistan Sum
WST - Tala
YER - Yemeni Rial
TWD - New Taiwan Dollar
SLE - Leone
VED - Bolívar Soberano
UYW - Unidad Previsional
VES - Bolívar Soberano
MRU - Ouguiya
STN - Dobra
CUC - Peso Convertible
ZWL - Zimbabwe Dollar
BYN - Belarusian Ruble
TMT - Turkmenistan New Manat
GHS - Ghana Cedi

SDG - Sudanese Pound
UYI - Uruguay Peso en Unidades Indexadas
RSD - Serbian Dinar
MZN - Mozambique Metical
AZN - Azerbaijan Manat
RON - Romanian Leu
CHE - WIR Euro
CHW - WIR Franc
TRY - Turkish Lira
XAF - CFA Franc BEAC
XCD - East Caribbean Dollar
XOF - CFA Franc BCEAO
XPF - CFP Franc
XBA - Bond Markets Unit European Composite Unit (EURCO)
XBB - Bond Markets Unit European Monetary Unit (E.M.U.-6)
XBC - Bond Markets Unit European Unit of Account 9 (E.U.A.-9)
XBD - Bond Markets Unit European Unit of Account 17 (E.U.A.-17)
XAU - Gold
XDR - SDR (Special Drawing Right)
XAG - Silver
XPT - Platinum
XTS - Codes specifically reserved for testing purposes
XPD - Palladium
XUA - ADB Unit of Account
ZMW - Zambian Kwacha
SRD - Surinam Dollar
MGA - Malagasy Ariary
COU - Unidad de Valor Real
AFN - Afghani
TJS - Somoni
AOA - Kwanza
BGN - Bulgarian Lev
CDF - Congolese Franc
BAM - Convertible Mark
EUR - Euro
MXV - Mexican Unidad de Inversion (UDI)
UAH - Hryvnia
GEL - Lari
BOV - Mvdol
PLN - Zloty
BRL - Brazilian Real
CLF - Unidades de fomento
XSU - Sucre

USN - US Dollar (Next day)

XXX - The codes assigned for transactions where no currency is involved

dateValidity (optional)

Long Date of validity of index format: int64

7.1.10. INDEXPORTFOLIOENTRYVIEW

Contains information about a product providing the index portfolio.

publicProductIdentification (optional)

PublicProductIdentificationView

mic (optional)

String Market structure's Market Identifier Code (MIC) as specified in ISO 10383.

currency (optional)

String Currency - ISO 4217 (2022/04/01) code (e.g. PLN)

Enum:

ALL - Lek
DZD - Algerian Dinar
ARS - Argentine Peso
AUD - Australian Dollar
BSD - Bahamian Dollar
BHD - Bahraini Dinar
BDT - Taka
AMD - Armenian Dram
BBD - Barbados Dollar
BMD - Bermudian Dollar
BTN - Ngultrum
BOB - Boliviano
BWP - Pula
BZD - Belize Dollar
SBD - Solomon Islands Dollar
BND - Brunei Dollar
MMK - Kyat
BIF - Burundi Franc
KHR - Riel
CAD - Canadian Dollar
CVE - Cabo Verde Escudo
KYD - Cayman Islands Dollar
LKR - Sri Lanka Rupee
CLP - Chilean Peso
CNY - Yuan Renminbi
COP - Colombian Peso
KMF - Comorian Franc
CRC - Costa Rican Colon

HRK - Kuna
CUP - Cuban Peso
CZK - Czech Koruna
DKK - Danish Krone
DOP - Dominican Peso
SVC - El Salvador Colon
ETB - Ethiopian Birr
ERN - Nakfa
FKP - Falkland Islands Pound
FJD - Fiji Dollar
DJF - Djibouti Franc
GMD - Dalasi
GIP - Gibraltar Pound
GTQ - Quetzal
GNF - Guinean Franc
GYD - Guyana Dollar
HTG - Gourde
HNL - Lempira
HKD - Hong Kong Dollar
HUF - Forint
ISK - Iceland Krona
INR - Indian Rupee
IDR - Rupiah
IRR - Iranian Rial
IQD - Iraqi Dinar
ILS - New Israeli Sheqel
JMD - Jamaican Dollar
JPY - Yen
KZT - Tenge
JOD - Jordanian Dinar
KES - Kenyan Shilling
KPW - North Korean Won
KRW - Won
KWD - Kuwaiti Dinar
KGS - Som
LAK - Lao Kip
LBP - Lebanese Pound
LSL - Loti
LRD - Liberian Dollar
LYD - Libyan Dinar
MOP - Pataca
MWK - Malawi Kwacha
MYR - Malaysian Ringgit

MVR - Rufiyaa
MUR - Mauritius Rupee
MXN - Mexican Peso
MNT - Tugrik
MDL - Moldovan Leu
MAD - Moroccan Dirham
OMR - Rial Omani
NAD - Namibia Dollar
NPR - Nepalese Rupee
ANG - Netherlands Antillean Guilder
AWG - Aruban Florin
VUV - Vatu
NZD - New Zealand Dollar
NIO - Cordoba Oro
NGN - Naira
NOK - Norwegian Krone
PKR - Pakistan Rupee
PAB - Balboa
PGK - Kina
PYG - Guarani
PEN - Sol
PHP - Philippine Peso
QAR - Qatari Rial
RUB - Russian Ruble
RWF - Rwanda Franc
SHP - Saint Helena Pound
SAR - Saudi Riyal
SCR - Seychelles Rupee
SGD - Singapore Dollar
VND - Dong
SOS - Somali Shilling
ZAR - Rand
SSP - South Sudanese Pound
SZL - Lilangeni
SEK - Swedish Krona
CHF - Swiss Franc
SYP - Syrian Pound
THB - Baht
TOP - Pa'anga
TTD - Trinidad and Tobago Dollar
AED - UAE Dirham
TND - Tunisian Dinar
UGX - Uganda Shilling

MKD - Denar
EGP - Egyptian Pound
GBP - Pound Sterling
TZS - Tanzanian Shilling
USD - US Dollar
UYU - Peso Uruguayo
UZS - Uzbekistan Sum
WST - Tala
YER - Yemeni Rial
TWD - New Taiwan Dollar
SLE - Leone
VED - Bolívar Soberano
UYW - Unidad Previsional
VES - Bolívar Soberano
MRU - Ouguiya
STN - Dobra
CUC - Peso Convertible
ZWL - Zimbabwe Dollar
BYN - Belarusian Ruble
TMT - Turkmenistan New Manat
GHS - Ghana Cedi
SDG - Sudanese Pound
UYI - Uruguay Peso en Unidades Indexadas
RSD - Serbian Dinar
MZN - Mozambique Metical
AZN - Azerbaijan Manat
RON - Romanian Leu
CHE - WIR Euro
CHW - WIR Franc
TRY - Turkish Lira
XAF - CFA Franc BEAC
XCD - East Caribbean Dollar
XOF - CFA Franc BCEAO
XPF - CFP Franc
XBA - Bond Markets Unit European Composite Unit (EURCO)
XBB - Bond Markets Unit European Monetary Unit (E.M.U.-6)
XBC - Bond Markets Unit European Unit of Account 9 (E.U.A.-9)
XBD - Bond Markets Unit European Unit of Account 17 (E.U.A.-17)
XAU - Gold
XDR - SDR (Special Drawing Right)
XAG - Silver
XPT - Platinum
XTS - Codes specifically reserved for testing purposes

XPD - Palladium
XUA - ADB Unit of Account
ZMW - Zambian Kwacha
SRD - Surinam Dollar
MGA - Malagasy Ariary
COU - Unidad de Valor Real
AFN - Afghani
TJS - Somoni
AOA - Kwanza
BGN - Bulgarian Lev
CDF - Congolese Franc
BAM - Convertible Mark
EUR - Euro
MXV - Mexican Unidad de Inversion (UDI)
UAH - Hryvnia
GEL - Lari
BOV - Mvdol
PLN - Zloty
BRL - Brazilian Real
CLF - Unidades de fomento
XSU - Sucre
USN - US Dollar (Next day)
XXX - The codes assigned for transactions where no currency is involved

instrumentPacket (optional)

Integer Instrument packet in index portfolio / of information product. Number of units (usually stocks/shares) of a given instrument in the index portfolio or in the information product.

7.1.11. INDEXREFERENCEDATAVIEW

Index metadata and portfolio

productName (optional)

String Product name.

params (optional)

IndexParamsView

portfolio (optional)

array[IndexPortfolioEntryView] Index portfolio.

7.1.12. INDEXSUMMARYVIEW

Summary statistics at the end of session for Index

instrumentId (optional)

Long ID of the instrument. format: int64

openingIndexValue (optional)

BigDecimal Opening index value.

sessionLow (optional)

BigDecimal Lowest value of the day.

sessionHigh (optional)

BigDecimal Highest value of the day.

closingIndexValue (optional)

BigDecimal Closing index value.

sessionAvg (optional)

BigDecimal Average value of the day.

pctChangeIndexValPrevSession (optional)

BigDecimal Percentage change of the index value in relation to the value of the closing index from the previous session.

pctChangeIndexValPrevYear (optional)

BigDecimal Percentage change of the index value in relation to its value at the end of the previous year.

closingCapitalisationPortfolio (optional)

BigDecimal Capitalisation of the portfolio for the closing index expressed in the currency of the index, on basis of which the closing index is calculated.

indIndexOpeningPortfolio (optional)

BigDecimal Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.

7.1.13. INSTRUMENTDETAILSVIEW

Detailed list of instruments.

tickTables (optional)

map[String, TickTableView] Tick Table map.

collars (optional)

map[String, CollarGroupView] Collar group map.

tradingSchedules (optional)

map[String, TradingScheduleView] Trading schedules map.

instruments (optional)

array[InstrumentView] List of Instrument Information.

7.1.14. INSTRUMENTKIDVIEW

Contains information about Key Information Detail for given instrument

productId (optional)

Long Product ID. format: int64

productIdentification (optional)

String Product identification, e.g. ISIN number.

productIdentificationType (optional)

String Type of product identification (for example ISIN or SEDOL).

Enum:

ISIN

UNKNOWN

url (optional)

String File URL where you can download KID

7.1.15. INSTRUMENTSIMPLIFIEDVIEW

Definition of a financial instrument.

instrumentId (optional)

Long ID of financial instrument. format: int64

code (optional)

String Financial instrument code.

productType (optional)

String Product Type

Enum:

FINANCIALPRODUCTSHARE
FINANCIALPRODUCTBOND
FINANCIALPRODUCTDERIVATIVEFUTURES
FINANCIALPRODUCTDERIVATIVEOPTIONS
FINANCIALPRODUCTINDEX
FINANCIALPRODUCTCURRENCY

~~UNKNOWN~~

productIdentification (optional)

String Product identification, e.g. ISIN number.

productIdentificationType (optional)

String Type of product identification (for example ISIN or SEDOL).

Enum:

ISIN

~~UNKNOWN~~

productName (optional)

String Name of the product.

7.1.16. INSTRUMENTSNAPSHOTVIEW

Extended instrument information for a given moment in time. Contains configuration information but also price information

timestamp (optional)

String Value in format yyyy-MM-dd HH:mm:ss, updated whenever any of fields change the value, this is a REST API internally generated value

referencePrice (optional)

BigDecimal Reference price from previous session.

instrumentId (optional)

Long ID of financial instrument. format: int64

code (optional)

String Financial instrument code.

productIdentification (optional)

String Product identification, e.g. ISIN number.

productIdentificationType (optional)

String Type of product identification (for example ISIN or SEDOL).

Enum:

ISIN

~~UNKNOWN~~

auctionUpdate (optional)

AuctionUpdateView

instrumentStatus (optional)

String Financial instrument status

Enum:

ACTIVE

INACTIVE

MARKETOPERATIONSSUSPENSION

OUTSIDECOLLARSSTATIC

OUTSIDECOLLARSDYNAMIC

REGULATORYSUSPENSION

TECHNICALHALT

HYBRIDNOQUOTES

HYBRIDPAUSE

~~UNKNOWN~~

tradingPhaseId (optional)

Long Trading phase id format: int64

tradingPhaseType (optional)

String Type of trading phase.

Enum:

CONTINUOUSPRICETIME

CONTINUOUSREFPRICETIME

CONTINUOUSLASTAUCTIONTIME

AUCTIONOPENING

BLOCKEXACTMATCHING

NOTRADINGCLOSED

UNCROSSING

AUCTIONCLOSING

AUCTIONINTRADAY

AUCTIONVOLATILITYSTATIC

AUCTIONVOLATILITYDYNAMIC

AUCTIONEXTENDEDVOLATILITYSTATIC

AUCTIONEXTENDEDVOLATILITYDYNAMIC

NOTRADINGEARLYMONITORING

NOTRADINGLATEMONITORING

HYBRID

HYBRIDBUYONLY

HYBRIDPRETRADE

UNSUSPENSIONAUCTION

IPO

~~UNKNOWN~~

TENDEROFFER

priceLevelSnapshot (optional)

PriceLevelSnapshotView

staticCollars (optional)

CollarsView

dynamicCollars (optional)

CollarsView

pctChange (optional)

BigDecimal Percentage change in price compared to the previous closing price.If previousClosingPrice is ZERO, it returns ZERO.

openPrice (optional)

BigDecimal Daily open price for given session. This is price taken from the first Trade message.

maxPrice (optional)

BigDecimal Daily max price for given session

minPrice (optional)

BigDecimal Daily min price for given session

vwap (optional)

BigDecimal Calculates the Volume Weighted Average Price (VWAP). VWAP is calculated as cumulativeValue to cumulativeVolume.

totalVolume (optional)

BigDecimal Total volume as a sum of all trades volume for given session

totalValue (optional)

BigDecimal Total value as a sum of all trades multiplied by their price and quantity for a given session

noTrades (optional)

Integer Total number of all trades for a given session format: int32

7.1.17. INSTRUMENTSUMMARYVIEW

Provides brief instrument summary.

lastTradedPrice (optional)

BigDecimal Last Traded Price (LTP).

closingPrice (optional)

BigDecimal Closing Price (CP).

closingPriceType (optional)

String Closing Price Type

Enum:

LTP

LASTACP

FAIRVALUE

DAILYSETTLEMENTPRICE

FINALSETTLEMENTPRICE

~~UNKNOWN~~

adjustedClosingPrice (optional)

BigDecimal Adjusted Closing Price (ACP).

adjustedClosingPriceReason (optional)

String Adjusted Closing Price Reason

Enum:

REGULAR

DIVIDEND

ISSUERIGHT

SPLIT

REVERSESPLIT

BONUS

SPINOFF

~~TICKTABLECHANGE~~

~~MULTIPLECA~~

~~UNKNOWN~~

pctChange (optional)

BigDecimal Percentage change.

noTrades (optional)

Integer Total number of transactions on the current trading day.

totalVolume (optional)

Integer Total transaction volume.

totalValue (optional)

BigDecimal Total transaction value.

openingPrice (optional)

BigDecimal The price of the first trade on the current trading day.

maxPrice (optional)

BigDecimal Highest price of the instrument on the current trading day.

minPrice (optional)

BigDecimal Lowest price of the instrument on the current trading day.

vwap (optional)

BigDecimal

7.1.18. INSTRUMENTVIEW

Extended Instrument Information

instrumentId (optional)

Long ID of financial instrument. format: int64

productType (optional)

String Product Type

Enum:

FINANCIALPRODUCTSHARE

FINANCIALPRODUCTBOND
FINANCIALPRODUCTDERIVATIVEFUTURES
FINANCIALPRODUCTDERIVATIVEOPTIONS
FINANCIALPRODUCTINDEX
FINANCIALPRODUCTCURRENCY

UNKNOWN

firstTradingDate (optional)

Long First trading day of the financial instrument. format: int64

lastTradingDate (optional)

Long Last trading day of the financial instrument. format: int64

productId (optional)

Long Reference to financial product definition. format: int64

currency (optional)

String Currency - ISO 4217 (2022/04/01) code (e.g. PLN)

Enum:

- ALL - Lek
- DZD - Algerian Dinar
- ARS - Argentine Peso
- AUD - Australian Dollar
- BSD - Bahamian Dollar
- BHD - Bahraini Dinar
- BDT - Taka
- AMD - Armenian Dram
- BBD - Barbados Dollar
- BMD - Bermudian Dollar
- BTN - Ngultrum
- BOB - Boliviano
- BWP - Pula
- BZD - Belize Dollar
- SBD - Solomon Islands Dollar
- BND - Brunei Dollar
- MMK - Kyat
- BIF - Burundi Franc
- KHR - Riel
- CAD - Canadian Dollar
- CVE - Cabo Verde Escudo
- KYD - Cayman Islands Dollar
- LKR - Sri Lanka Rupee
- CLP - Chilean Peso
- CNY - Yuan Renminbi
- COP - Colombian Peso
- KMF - Comorian Franc
- CRC - Costa Rican Colon

HRK - Kuna
CUP - Cuban Peso
CZK - Czech Koruna
DKK - Danish Krone
DOP - Dominican Peso
SVC - El Salvador Colon
ETB - Ethiopian Birr
ERN - Nakfa
FKP - Falkland Islands Pound
FJD - Fiji Dollar
DJF - Djibouti Franc
GMD - Dalasi
GIP - Gibraltar Pound
GTQ - Quetzal
GNF - Guinean Franc
GYD - Guyana Dollar
HTG - Gourde
HNL - Lempira
HKD - Hong Kong Dollar
HUF - Forint
ISK - Iceland Krona
INR - Indian Rupee
IDR - Rupiah
IRR - Iranian Rial
IQD - Iraqi Dinar
ILS - New Israeli Sheqel
JMD - Jamaican Dollar
JPY - Yen
KZT - Tenge
JOD - Jordanian Dinar
KES - Kenyan Shilling
KPW - North Korean Won
KRW - Won
KWD - Kuwaiti Dinar
KGS - Som
LAK - Lao Kip
LBP - Lebanese Pound
LSL - Loti
LRD - Liberian Dollar
LYD - Libyan Dinar
MOP - Pataca
MWK - Malawi Kwacha
MYR - Malaysian Ringgit

MVR - Rufiyaa
MUR - Mauritius Rupee
MXN - Mexican Peso
MNT - Tugrik
MDL - Moldovan Leu
MAD - Moroccan Dirham
OMR - Rial Omani
NAD - Namibia Dollar
NPR - Nepalese Rupee
ANG - Netherlands Antillean Guilder
AWG - Aruban Florin
VUV - Vatu
NZD - New Zealand Dollar
NIO - Cordoba Oro
NGN - Naira
NOK - Norwegian Krone
PKR - Pakistan Rupee
PAB - Balboa
PGK - Kina
PYG - Guarani
PEN - Sol
PHP - Philippine Peso
QAR - Qatari Rial
RUB - Russian Ruble
RWF - Rwanda Franc
SHP - Saint Helena Pound
SAR - Saudi Riyal
SCR - Seychelles Rupee
SGD - Singapore Dollar
VND - Dong
SOS - Somali Shilling
ZAR - Rand
SSP - South Sudanese Pound
SZL - Lilangeni
SEK - Swedish Krona
CHF - Swiss Franc
SYP - Syrian Pound
THB - Baht
TOP - Pa'anga
TTD - Trinidad and Tobago Dollar
AED - UAE Dirham
TND - Tunisian Dinar
UGX - Uganda Shilling

MKD - Denar
EGP - Egyptian Pound
GBP - Pound Sterling
TZS - Tanzanian Shilling
USD - US Dollar
UYU - Peso Uruguayo
UZS - Uzbekistan Sum
WST - Tala
YER - Yemeni Rial
TWD - New Taiwan Dollar
SLE - Leone
VED - Bolívar Soberano
UYW - Unidad Previsional
VES - Bolívar Soberano
MRU - Ouguiya
STN - Dobra
CUC - Peso Convertible
ZWL - Zimbabwe Dollar
BYN - Belarusian Ruble
TMT - Turkmenistan New Manat
GHS - Ghana Cedi
SDG - Sudanese Pound
UYI - Uruguay Peso en Unidades Indexadas
RSD - Serbian Dinar
MZN - Mozambique Metical
AZN - Azerbaijan Manat
RON - Romanian Leu
CHE - WIR Euro
CHW - WIR Franc
TRY - Turkish Lira
XAF - CFA Franc BEAC
XCD - East Caribbean Dollar
XOF - CFA Franc BCEAO
XPF - CFP Franc
XBA - Bond Markets Unit European Composite Unit (EURCO)
XBB - Bond Markets Unit European Monetary Unit (E.M.U.-6)
XBC - Bond Markets Unit European Unit of Account 9 (E.U.A.-9)
XBD - Bond Markets Unit European Unit of Account 17 (E.U.A.-17)
XAU - Gold
XDR - SDR (Special Drawing Right)
XAG - Silver
XPT - Platinum
XTS - Codes specifically reserved for testing purposes

- XPD - Palladium
- XUA - ADB Unit of Account
- ZMW - Zambian Kwacha
- SRD - Surinam Dollar
- MGA - Malagasy Ariary
- COU - Unidad de Valor Real
- AFN - Afghani
- TJS - Somoni
- AOA - Kwanza
- BGN - Bulgarian Lev
- CDF - Congolese Franc
- BAM - Convertible Mark
- EUR - Euro
- MXV - Mexican Unidad de Inversion (UDI)
- UAH - Hryvnia
- GEL - Lari
- BOV - Mvdol
- PLN - Zloty
- BRL - Brazilian Real
- CLF - Unidades de fomento
- XSU - Sucre
- USN - US Dollar (Next day)
- XXX - The codes assigned for transactions where no currency is involved

lotSize (optional)

Long Lot size for the instrument. format: int64

priceExpressionType (optional)

String Price expression type for the financial instrument

Enum:

- PRICE
- PERCENTAGE
- ~~UNKNOWN~~

calendarId (optional)

Long ID of trading calendar for the financial instrument. format: int64

marketStructureId (optional)

Long ID of the financial instrument's market segment. format: int64

tickTableId (optional)

Long TickTable ID for the financial instrument. format: int64

collarGroupId (optional)

Long Collar group ID. format: int64

tradingScheduleId (optional)

Long Trading schedule ID. format: int64

code (optional)

String Financial instrument code.

mic (optional)

String Market structure's Market Identifier Code (MIC) as specified in ISO 10383.

nominalValue (optional)

BigDecimal Nominal value of the financial instrument.

nominalValueType (optional)

String Type of the product's nominal value (no nominal, constant or unknown)

Enum:

NOTAPPLICABLE

CONSTANT

~~UNKNOWN~~

multiplier (optional)

BigDecimal Product value multiplier.

strikePrice (optional)

BigDecimal Product strike price.

productIdentification (optional)

String Product identification, e.g. ISIN number.

productIdentificationType (optional)

String Type of product identification (for example ISIN or SEDOL).

Enum:

ISIN

~~UNKNOWN~~

settlementCalendarId (optional)

Long Settlement calendar ID. format: int64

bondCouponType (optional)

String A bond's coupon type

Enum:

NOTAPPLICABLE

ZERO

FIXED

FLOATING

INDEXED

~~UNKNOWN~~

preTradeCheckMinPrice (optional)

BigDecimal Minimum order price for pre-trade check.

preTradeCheckMaxPrice (optional)

BigDecimal Maximum order price for pre-trade check.

preTradeCheckMinQuantity (optional)

Integer Minimum volume for pre-trade check.

preTradeCheckMaxQuantity (optional)

Integer Maximum volume for pre-trade check.

preTradeCheckMaxValue (optional)

BigDecimal Maximum value for pre-trade check.

preTradeCheckMinValue (optional)

BigDecimal Minimum value for pre-trade check.

liquidity (optional)

Boolean Liquidity flag, true - liquid, false - illiquid.

marketModelType (optional)

String Market Model

Enum:

NOTAPPLICABLE

CLOB

BLOCK

HYBRID

CROSS

IPO

~~UNKNOWN~~

issuer (optional)

String Name of the issuer.

issuerRegCountry (optional)

String Country codes based on ISO 3166.

Enum:

AUS - Australia

AUT - Austria

BEL - Belgium

BGR - Bulgaria

CAN - Canada

CYP - Cyprus

CZE - Czechia

EST - Estonia

FRA - France

DEU - Germany

HUN - Hungary

IRL - Ireland

ISR - Israel

ITA - Italy

LTU - Lithuania

LUX - Luxembourg

NLD - Netherlands (the)

POL - Poland

PRT - Portugal

ROU - Romania

SVK - Slovakia

SVN - Slovenia

ESP - Spain

SWE - Sweden

UKR - Ukraine

GBR - United Kingdom of Great Britain and Northern Ireland (the)

USA - United States of America (the)

underlyingInstrumentId (optional)

Long Identifier of the underlying instrument. format: int64

productCode (optional)

String Unique code of the product.

cfi (optional)

String CFI (Classification of Financial Instrument) code.

fisn (optional)

String Product FISN (Financial Instrument Short Name) code.

issueSize (optional)

Integer Size of the issue (depends on the issue size type).

nominalCurrency (optional)

String Currency - ISO 4217 (2022/04/01) code (e.g. PLN)

Enum:

ALL - Lek

DZD - Algerian Dinar

ARS - Argentine Peso

AUD - Australian Dollar

BSD - Bahamian Dollar

BHD - Bahraini Dinar

BDT - Taka

AMD - Armenian Dram

BBD - Barbados Dollar

BMD - Bermudian Dollar

BTN - Ngultrum

BOB - Boliviano

BWP - Pula

BZD - Belize Dollar

SBD - Solomon Islands Dollar

BND - Brunei Dollar

MMK - Kyat

BIF - Burundi Franc

KHR - Riel

CAD - Canadian Dollar

CVE - Cabo Verde Escudo

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LKR - Sri Lanka Rupee

CLP - Chilean Peso

CNY - Yuan Renminbi

COP - Colombian Peso

KMF - Comorian Franc

CRC - Costa Rican Colon

HRK - Kuna
CUP - Cuban Peso
CZK - Czech Koruna
DKK - Danish Krone
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FKP - Falkland Islands Pound
FJD - Fiji Dollar
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GMD - Dalasi
GIP - Gibraltar Pound
GTQ - Quetzal
GNF - Guinean Franc
GYD - Guyana Dollar
HTG - Gourde
HNL - Lempira
HKD - Hong Kong Dollar
HUF - Forint
ISK - Iceland Krona
INR - Indian Rupee
IDR - Rupiah
IRR - Iranian Rial
IQD - Iraqi Dinar
ILS - New Israeli Sheqel
JMD - Jamaican Dollar
JPY - Yen
KZT - Tenge
JOD - Jordanian Dinar
KES - Kenyan Shilling
KPW - North Korean Won
KRW - Won
KWD - Kuwaiti Dinar
KGS - Som
LAK - Lao Kip
LBP - Lebanese Pound
LSL - Loti
LRD - Liberian Dollar
LYD - Libyan Dinar
MOP - Pataca
MWK - Malawi Kwacha
MYR - Malaysian Ringgit

MVR - Rufiyaa
MUR - Mauritius Rupee
MXN - Mexican Peso
MNT - Tugrik
MDL - Moldovan Leu
MAD - Moroccan Dirham
OMR - Rial Omani
NAD - Namibia Dollar
NPR - Nepalese Rupee
ANG - Netherlands Antillean Guilder
AWG - Aruban Florin
VUV - Vatu
NZD - New Zealand Dollar
NIO - Cordoba Oro
NGN - Naira
NOK - Norwegian Krone
PKR - Pakistan Rupee
PAB - Balboa
PGK - Kina
PYG - Guarani
PEN - Sol
PHP - Philippine Peso
QAR - Qatari Rial
RUB - Russian Ruble
RWF - Rwanda Franc
SHP - Saint Helena Pound
SAR - Saudi Riyal
SCR - Seychelles Rupee
SGD - Singapore Dollar
VND - Dong
SOS - Somali Shilling
ZAR - Rand
SSP - South Sudanese Pound
SZL - Lilangeni
SEK - Swedish Krona
CHF - Swiss Franc
SYP - Syrian Pound
THB - Baht
TOP - Pa'anga
TTD - Trinidad and Tobago Dollar
AED - UAE Dirham
TND - Tunisian Dinar
UGX - Uganda Shilling

MKD - Denar
EGP - Egyptian Pound
GBP - Pound Sterling
TZS - Tanzanian Shilling
USD - US Dollar
UYU - Peso Uruguayo
UZS - Uzbekistan Sum
WST - Tala
YER - Yemeni Rial
TWD - New Taiwan Dollar
SLE - Leone
VED - Bolívar Soberano
UYW - Unidad Previsional
VES - Bolívar Soberano
MRU - Ouguiya
STN - Dobra
CUC - Peso Convertible
ZWL - Zimbabwe Dollar
BYN - Belarusian Ruble
TMT - Turkmenistan New Manat
GHS - Ghana Cedi
SDG - Sudanese Pound
UYI - Uruguay Peso en Unidades Indexadas
RSD - Serbian Dinar
MZN - Mozambique Metical
AZN - Azerbaijan Manat
RON - Romanian Leu
CHE - WIR Euro
CHW - WIR Franc
TRY - Turkish Lira
XAF - CFA Franc BEAC
XCD - East Caribbean Dollar
XOF - CFA Franc BCEAO
XPF - CFP Franc
XBA - Bond Markets Unit European Composite Unit (EURCO)
XBB - Bond Markets Unit European Monetary Unit (E.M.U.-6)
XBC - Bond Markets Unit European Unit of Account 9 (E.U.A.-9)
XBD - Bond Markets Unit European Unit of Account 17 (E.U.A.-17)
XAU - Gold
XDR - SDR (Special Drawing Right)
XAG - Silver
XPT - Platinum
XTS - Codes specifically reserved for testing purposes

XPD - Palladium
XUA - ADB Unit of Account
ZMW - Zambian Kwacha
SRD - Surinam Dollar
MGA - Malagasy Ariary
COU - Unidad de Valor Real
AFN - Afghani
TJS - Somoni
AOA - Kwanza
BGN - Bulgarian Lev
CDF - Congolese Franc
BAM - Convertible Mark
EUR - Euro
MXV - Mexican Unidad de Inversion (UDI)
UAH - Hryvnia
GEL - Lari
BOV - Mvdol
PLN - Zloty
BRL - Brazilian Real
CLF - Unidades de fomento
XSU - Sucre
USN - US Dollar (Next day)
XXX - The codes assigned for transactions where no currency is involved

usIndicator (optional)

String US Regulation S indicator.

Enum:

- NOTAPPLICABLE
- REGULATIONS
- REGULATIONSPUSRULE144A
- ~~UNKNOWN~~

expiryDate (optional)

Long Expiration date. format: int64

versionNumber (optional)

Integer Version of the derivative. Version is incremented when corporate event adjusting underlying closing price occurs. format: int32

settlementType (optional)

String Indicates settlement type of the derivative.

Enum:

- NOTAPPLICABLE
- CASHSETTLEMENT
- PHYSICALDELIVERY
- ~~UNKNOWN~~

optionType (optional)

String Type of option.

Enum:

NOTAPPLICABLE

CALL

PUT

~~UNKNOWN~~

exerciseType (optional)

String Dictionary with option exercise styles

Enum:

AMER

EURO

NA

~~UNKNOWN~~

productName (optional)

String Name of the product.

referencePrice (optional)

BigDecimal Reference price.

status (optional)

String Financial instrument status

Enum:

ACTIVE

INACTIVE

MARKETOPERATIONSSUSPENSION

OUTSIDECOLLARSSTATIC

OUTSIDECOLLARSDYNAMIC

REGULATORYSUSPENSION

TECHNICALHALT

HYBRIDNOQUOTES

HYBRIDPAUSE

~~UNKNOWN~~

initialPhaseId (optional)

Long Id of starting phase for the instrument. format: int64

7.1.19. MARKETSTRUCTUREVIEW

Messages defining the Trading Venue's market structure.

id (optional)

Long ID of the market structure. format: int64

mic (optional)

String Market structure's Market Identifier Code (MIC) as specified in ISO 10383.

name (optional)

String Market structure name assigned by market operator.

marketModelType (optional)

String Market Model

Enum:

NOTAPPLICABLE

CLOB

BLOCK

HYBRID

CROSS

IPO

~~UNKNOWN~~

children (optional)

array[MarketStructureView] Nested market structures.

7.1.20. PRICELEVELSNAPSHOTVIEW

BBO price levels and volumes

maxDepth (optional)

Integer The number of BBO levels contained in the message. format: int32

buy (optional)

array[PriceLevelView] Price levels for buy side.

sell (optional)

array[PriceLevelView] Price levels for sell side.

7.1.21. PRICELEVELVIEW

Data type for storing snapshot price level.

price (optional)

BigDecimal Price level in currency units.

quantity (optional)

Integer Quantity of securities available at a given price level.

orderCount (optional)

Integer Number of open orders at a given price level. format: int32

7.1.22. PUBLICPRODUCTIDENTIFICATIONVIEW

Product identification based on well-known identification types, such as ISIN.

productIdentification (optional)

String Product identification, for example its ISIN code.

productIdentificationType (optional)

String Type of product identification (for example ISIN or SEDOL).

Enum:

ISIN

~~UNKNOWN~~

7.1.23. REALTIMEINDEXVIEW

Message providing real-time index values for instruments.

instrumentId (optional)

Long ID of the instrument. format: int64

indexLevelCode (optional)

String Enumeration of index levels

Enum:

PREVIOUSCLOSINGVALUE

THEORETICALVALUEBEFOREOPENING

VALUEBEFOREOPENING

OPENINGVALUE

CURRENTVALUE

PRECLOSINGVALUE

CLOSINGVALUE

VALUEWITHOUTOPENING

~~UNKNOWN~~

numOfActiveInstruments (optional)

Integer Number of instruments – participants of the index, for which the quotation was set during the session. format: int32

indIndexOpeningPortfolio (optional)

BigDecimal Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.

indexValue (optional)

BigDecimal The value of the last level for the index according to the definition provided in the indexLevelCode.

pctChangeIndexValPrevSession (optional)

BigDecimal Percentage change of the index value in relation to the value of the closing index from the previous session.

tradingValue (optional)

BigDecimal Value of trading in the instruments from the given index.

sessionLow (optional)

BigDecimal Lowest value of the day.

sessionHigh (optional)

BigDecimal Highest value of the day.

indexValueBid (optional)

BigDecimal Value of the buy index based on the best sell offers of instruments.

indexValueAsk (optional)

BigDecimal Value of the sell index based on the best sell offers of instruments.

midSpreadIndex (optional)

BigDecimal Mean of indexValueBid and indexValueAsk.

differenceCentralSpread (optional)

BigDecimal The point difference between midSpreadIndex and indexValue.

7.1.24. TICKTABLEVIEW

Tick Table.

tickTableId (optional)

Long Tick table ID. format: int64

ticks (optional)

array[TickView] Ticks.

7.1.25. TICKVIEW

Tick size definition.

tickSize (optional)

Long Size of the tick. format: int64

lowerBound (optional)

BigDecimal Tick size lower bound.

7.1.26. TRADINGPHASEVIEW

Trading phase definition

tradingPhaseId (optional)

Long Trading phase ID. format: int64

tradingPhaseSettlementCycle (optional)

Integer Settlement cycle for transactions made during a continuous phase. format: int32

tradingPhaseMaxBlockSettlementCycle (optional)

Integer Max settlement cycle for block transactions. format: int32

staticCollarVolatilityAuctionId (optional)

Long Static collar volatility auction ID. format: int64

dynamicCollarVolatilityAuctionId (optional)

Long Dynamic collar volatility auction ID. format: int64

tradingPhaseStartTime (optional)

Integer Trading phase start time.

tradingPhaseType (optional)

String Type of trading phase.

Enum:

CONTINUOUSPRICETIME
CONTINUOUSREFPRICETIME
CONTINUOUSLASTAUCTIONTIME
AUCTIONOPENING
BLOCKEXACTMATCHING
NOTRADINGCLOSED
UNCROSSING
AUCTIONCLOSING
AUCTIONINTRADAY
AUCTIONVOLATILITYSTATIC

AUCTIONVOLATILITYDYNAMIC
AUCTIONEXTENDEDVOLATILITYSTATIC
AUCTIONEXTENDEDVOLATILITYDYNAMIC
NOTRADINGEARLYMONITORING
NOTRADINGLATEMONITORING
HYBRID
HYBRIDBUYONLY
HYBRIDPRETRADE
UNSUSPENSIONAUCTION
IPO

~~UNKNOWN~~

auctionType (optional)

String Type of auction

Enum:

NOTAPPLICABLE
OPENINGAUCTION
CLOSINGAUCTION
INTRADAYAUCION
VOLATILITYAUCTIONSTATIC
VOLATILITYAUCTIONDYNAMIC
EXTENDEDVOLATILITYAUCTIONSTATIC
EXTENDEDVOLATILITYAUCTIONDYNAMIC
UNSUSPENSIONAUCTION

~~UNKNOWN~~

uncrossing (optional)

Boolean True if the phase includes an uncrossing.

extStaticCollarVolatilityAuctionId (optional)

Long ID of Static Collar Volatility Auction. format: int64

extDynamicCollarVolatilityAuctionId (optional)

Long ID of Dynamic Collar Volatility Auction. format: int64

lastAuctionPhaseId (optional)

Long ID of last auction phase. format: int64

7.1.27. TRADINGSCHEDULEVIEW

Trading schedule

tradingScheduleId (optional)

Long Trading schedule ID. format: int64

phases (optional)

array[TradingPhaseView] Trading phases.